Adam Bawatneh

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Research Interests

Social reasoning and Theory of Mind, including how people represent others' beliefs, intentions, and interactions.
Natural language processing, model evaluation, and responsible AI; current social reasoning research uses LLMs.

Education

University of Central Florida Orlando, Florida
Ph.D. in Computer Science Fall 2025 – Present; GPA: 4.00/4.00
University of Central Florida Orlando, Florida
B.S. in Computer Science Spring 2025

Selected Publication

Adam Bawatneh (sole first author), Sagar Sapkota, Amrit Singh Bedi, Santu Karmaker, and Mubarak Shah.
"OmniToM: Benchmarking Theory of Mind in LLMs via Explicit Belief Modeling."
NeurIPS 2026, accepted, Evaluations & Datasets Track. Project | Benchmark

Research Experience

Graduate Research Assistant, University of Central Florida May 2026 – Present
Conduct AI research and contribute to government proposal development, including IARPA-focused proposals.
Undergraduate Teaching Assistant, University of Central Florida May 2023 – Spring 2025
Led labs and mentored students in C, Python, and Java courses serving more than 1,500 students.

Selected Research Projects

OmniToM: Social Reasoning Evaluation for LLMs Oct 2025 – Present

- Co-developed a benchmark with 895 stories, 22,343 belief propositions, and 156,401 labels; coordinated annotation quality control across 11 annotators and about 1.1K person-hours.
- Built structured extraction and labeling prompts; evaluated API and open-weight models across Azure OpenAI, Anthropic, Vertex AI, and university/NSF ACCESS HPC.

CADS: Context-Aware Detection of Specialized Language Summer 2026 – Present
Mentored Valerie Lopez during the summer REU; the project continues in the lab.

- Developed a pipeline combining translation instability, surface clues, masked-language-model likelihood, and model agreement, with logistic-regression fusion.
- Reported 80.1 F1 on GenZ-A; broader domain and multilingual tests exposed limits in transfer.

BIAS-101: Demographic Bias in Video Action Recognition Jan 2025 – Apr 2025
Processed 85,000+ candidate videos and added 2,000+ curated clips; used multimodal annotation and statistical audits to study UCF-101 representation.
DETECT: Real-Time Gaze and Facial-Signal Analysis Aug 2024 – Apr 2025
Led a 6-member team; built a MediaPipe FaceMesh pipeline and deployed an Astro, Go, PostgreSQL, and Docker application. Project evaluation reported 80%+ gaze-tracking accuracy.

Academic Service

Conference reviewer: NeurIPS, AAAI, and ACL.

Technical Skills

AI and evaluation: Social reasoning, Theory of Mind, LLM evaluation, benchmark design and QA
NLP: Machine translation, text classification, masked-LM scoring, named entity recognition, fuzzy matching, model fusion
Programming: Python, C/C++, Java, JavaScript, Go, SQL
Platforms: Azure OpenAI, Anthropic API, Google Vertex AI, Hugging Face, Azure, Google Cloud, HPC
Tools: Git/GitHub, Linux, Docker, PostgreSQL, REST APIs, LaTeX, Jira